National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Embedded Systems Lab	Course Code:	EL419
Program:	BS(EE)	Semester:	Fall 2018
Duration:	90 Minutes	Total Marks:	50
Paper Date:	12-12-2017	Weightage	35%
Section:	ALL	Page(s):	2
Exam Type:	Final		

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Instruction/Notes:OPEN BOOK/NOTES

PART-A: Theoritical Question: [20 Marks] *Write answers of this question on the same sheet. No need of practical implementation of this part.*

1. **A temperature sensor is connected with one of the analog channels of ADC module of TM4C. Sensor outputs voltage from 0 to 3.3 volts. Assume we have done all the initiallizations of ADC module 0 channel 5. Answer the following questions: [10 Marks]**

   A. To which pin of TM4C you will connect your temperature sensor: Pin #63 [2]
   B. How to select channel 5: ______ [2]
   C. Write a piece of code to set a boolean variable test to 1 if output of temperature sensor is greater than 2.5 voltage and set test to 0 if it is less than 2.5 voltage. [6]

B) channel 5 is connected with ~~[illegible]~~
Pin 2 of GPIO Port D.
GPIO_PORTD_AFSEL_R |= 4;
GPIO_PORTD_DEN_R &= ~4;
GPIO_PORTD_AMSEL_R |= 4;
ADC0_SSMUX3_R = 0;

```
float fvin;
boolean test;
C) while(1)
{ ADC0_PSSI_R |= 0x0008;   // sample sequencer 3
while ((ADC0_RIS_R & 0x08);
fvin = ((ADC0_SSFIO3_R & 0xFFF)* 3.3)/4096;
ADC0_ISC_R = 8;
if (fvin >= 2.5)
{ test = 1 }
else test = 0
}
```